

Environmental Impact Analysis for Holistic Health Revenue Optimization Summary

This project analyzed the relationship between environmental factors(especially pollen levels and weather conditions) and healthcare services demand for a holistic health clinic offering respiratory, immunity, and wellness services. A publicly available grass pollen dataset from kaggle is combined with a synthetically generated clinical dataset to simulate patient visits and revenue patterns over time.

To investigate if the spikes in the air pollution or pollen count have any relation with the respiratory(allergy, athma) and immunity of people and to justify the increasing budget during this time.

Data Preparation and Inspection:

The grass pollen dataset from kaggle is used, which has 1953 rows with 37 columns. However, it was hard to find the respiratory dataset from kaggle and other sources. For this project generating synthetic data is an option, so I prefer to generate synthetic data.

Before generating synthetic data, the pollen index dataset was cleaned.

It had multiple irrelevant columns, null values, no duplicate values. 25 columns dropped, since the null values on those columns were also removed.

The date d-type changed from object to datetime.

The date is extracted yearly and monthly for future time series analysis.

And the project needs environmental stressors -> clinical demand-> Revenue impact, so before generating the clinical synthetic data. We select some columns that clinical data depends on

Pollen_count increase, -> respiratory_visits increase

Humidity increase, -> respiratory_visits increase

Pollen_count increase, -> symptoms worse

Wind_speed increase, -> pollen spread increases

Rain reduces pollen impact

High dew point means there is more moisture in air, so pollen can stick longer in the air this can worsen allergy symptoms

Hypothesis 1: Pollen_count has a direct cause of allergies.

So we need to see on this project if the higher pollen causes more respiratory issues, and it has relation with clinic revenue. For this we need evidence. That's why we need to analyse the data and interpret it based on the evidence we have.

So let us see what the columns we selected mean:

Pollen_cat: categorical pollen level: which has low, moderate, high and very high categories which helps for example revenue is 30% increase during high pollen days

Humidity: for example high humidity can trap pollen near ground, worsen breathing conditions. Can be a strong predictor of clinic visits

Rain: rain reduces airborne pollen temporarily but after rain plants release more pollen

Wind: wind_speed spreads pollen across regions, increase exposure

Season: Categorical grouping of months

Example: Spring, Summer, Autumn, Winter

For example we can say demand peaks in spring season

Now how we created the synthetic data

We created four additional columns on the existing 12 columns pollen dataset. The four columns created are:

Respiratory_visits, immunity_visits, wellness_visits and revenue

We used these baselines for immunity_visits

Pollen(mild effect), humidity(weak positive effect: people feel sick in humidity air), wind(small effect)

For wellness_visits we used np.sin, this creates seasonality pattern peaks in some months and dips in others

On revenue we converted visits into money. Respiratory has high value estimated \$120 per visit, immunity \$100 and \$80 for check up / wellness_visit

So, after generating synthetic data we have 1953 rows with 16 columns of clean dataset. Which can show as the relation of the environmental stress, clinical demand and budget.

Exploratory Data Analysis

Time Series analysis which helps to understand how environmental factors affect clinical demand.

From the analysis of pollen vs respiratory visits:

Respiratory visits closely follow pollen trends over time like when pollen increases visits increase when pollen is stable visits are also stable.

However when we see seasonal behaviour between 2011 and 2015, recurring peaks suggest pollen cycles that consistently drive clinical demand.

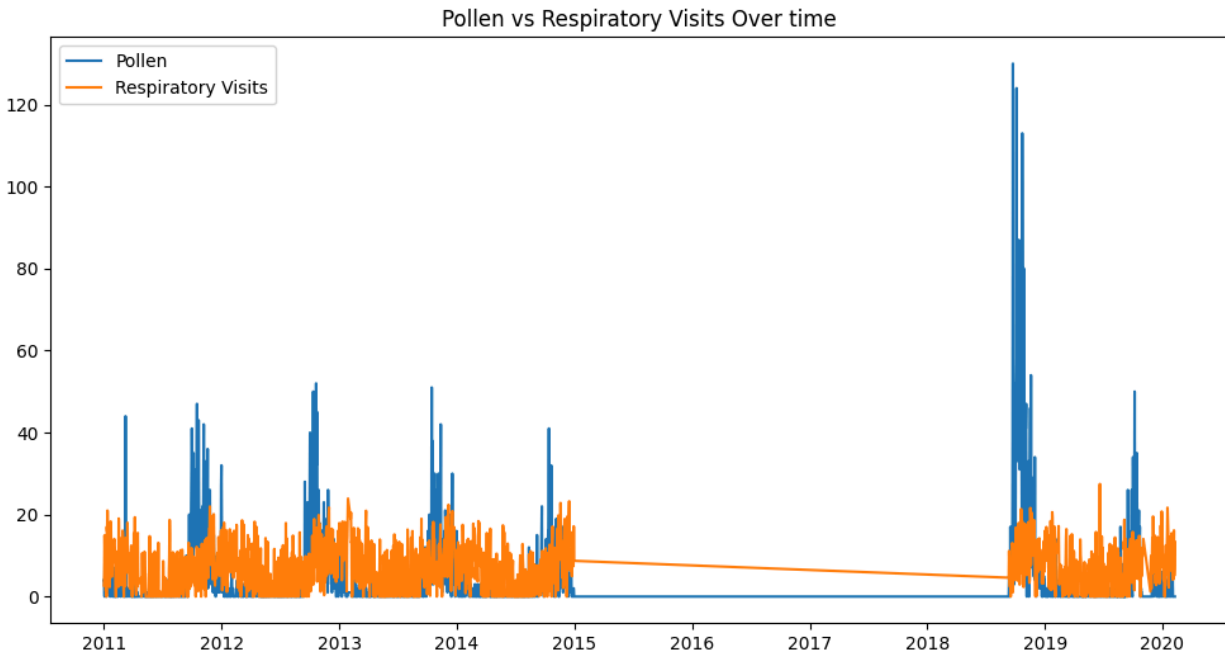
A flat period(2015 -2019) possible interpretation might be a period of relatively stable pollen levels resulting in consistent, lower variability in respiratory visits.

Almost no spikes, less urgent demand, stable revenue period.

Major spike around 2019 a sharp spike in pollen levels corresponds to a sharp increase in pollen levels doesn't always result in an equivalent increase in clinic visits.

Lag effect people don't go to the clinic immediately symptoms appear after exposure or people medicate themselves in their home.

Despite this, the overall trend alignment across the dataset still supports a positive relationship between environmental stressors and patient demand.



Correlation Analysis:

Check the correlation between, some important elements

pollen-> respiratory_visits ~ 0.19

pollen->revenue = 0.126963

weather(wind_speed, rain, humidity)->respiratory_visits

wind_speed-> respiratory_visits = 0.854947

humidity-> respiratory_visits = -0.477160

rain-> respiratory_visits = -0.203767

respiratory_visits->revenue = 0.889470

wind_speed -> revenue = 0.915284

Pollen has a positive but moderate impact on respiratory visits(0.19 corr), not every every pollen spike -> visit spike

Wind speed vs respiratory visits ~ 0.85, wind_speed shows a very strong positive correlation with respiratory visits, suggesting that airborne pollen dispersion plays a critical role in triggering patient demand.

Pollen_level vs respiratory_visits ~ 0.28

Categorical pollen levels better capture patient response than low pollen counts.

respiratory_visits->revenue ~ 0.89 and wind_speed -> revenue = 0.915284

Are the main revenue drivers

Wind_speed indirectly drives revenue by increasing respiratory related demand.

Pollen_count vs revenue ~ 0.12

Pollen does not directly drive revenue strongly, but influences it indirectly through respiratory visits

Environmental conditions that influence pollen dispersion(such as wind_speed) are stronger predictors of clinic demand than pollen_levels alone. Additionally, humidity shows negative correlation with respiratory demand ($r = -0.47$), suggesting that moist conditions may suppress airborne allergens.

Seasonal Trend:

In season: ~ 9.75

Not in season: ~ 0.80 that is more than 10x higher pollen. Pollen exposure is strongly seasonal and higher concentration in specific periods.

Clinic visits increase in season In season 8.87 than visits not In season: 5.33 visits. Respiratory demand increases significantly during high pollen seasons revenue also increases in season In season: 4252 and 3896 in Not In season.

High pollen seasons lead to a substantial increase in clinic visits, particularly respiratory related consultations. However, revenue growth is more moderate suggesting that while demand increases significantly, pricing structure or service mix may dilute overall revenue impact.

Lag Analysis:

pollen_lag1 vs respiratory_visits ~ 0.19 , pollen from the previous day has a small positive effect on respiratory visits today.

It may show that people do not react instantly to pollen exposure. They react after symptoms develop after irritation builds up sometimes with delay(1-2 days)

Respiratory demand is not strongly immediate but shows mild delayed response to environmental exposure. Environmental impact on healthcare demand is gradual, not immediate.

Pollen_cat & Revenue Impact:

Revenue increases with pollen level

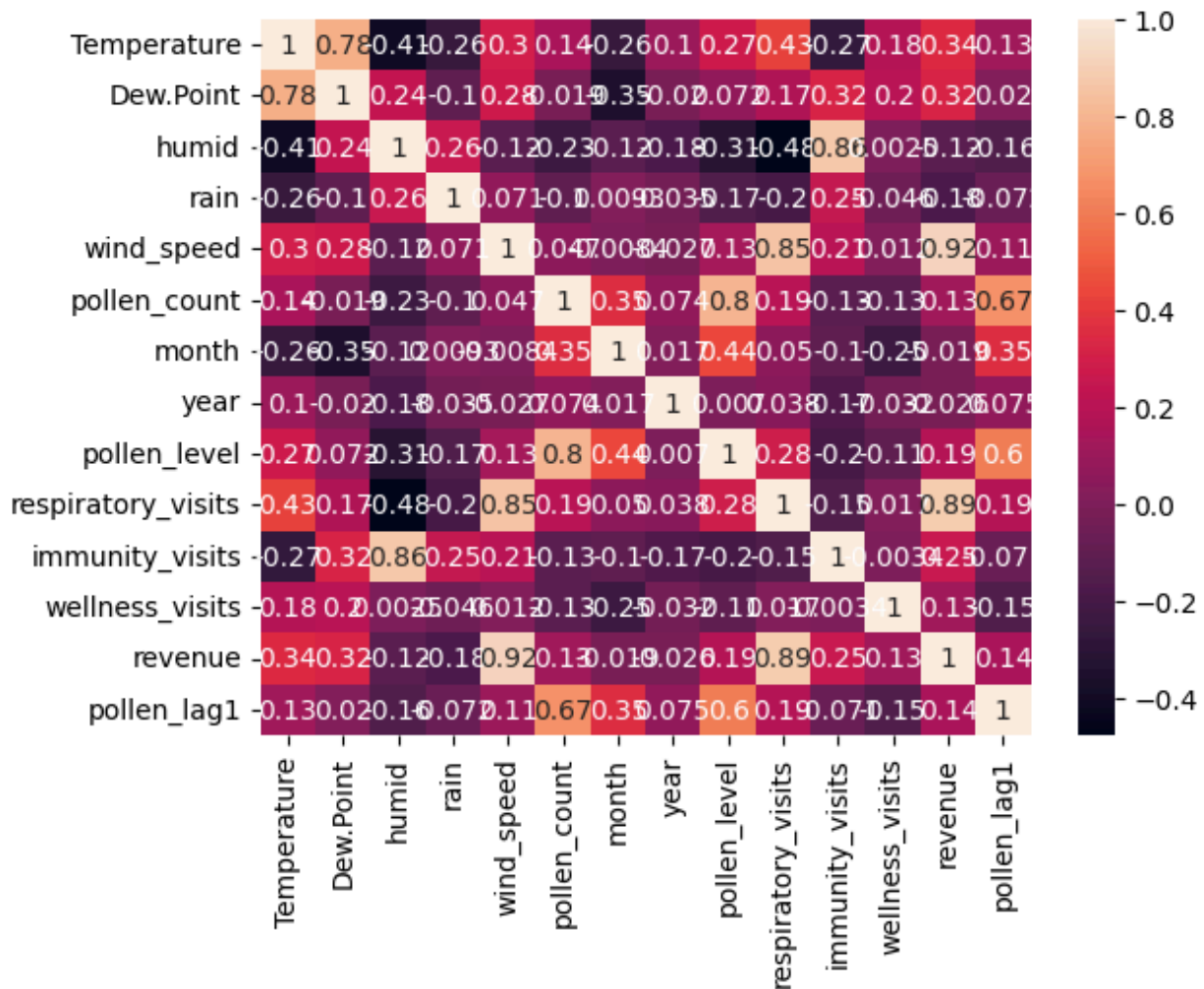
Very low-> lowest revenue(3919)

Very high->highest revenue(4359)

$4359 - 3919 = 440$ increase, 11% revenue increase

Revenue increases consistently as pollen intensity rises, indicating strong environmental sensitivity in clinic demand

More pollen ->more allergies->more clinic visits-> more revenue
relationship is steady increase.



Insights from the above heatmap graph:

Which have strong positive correlation?

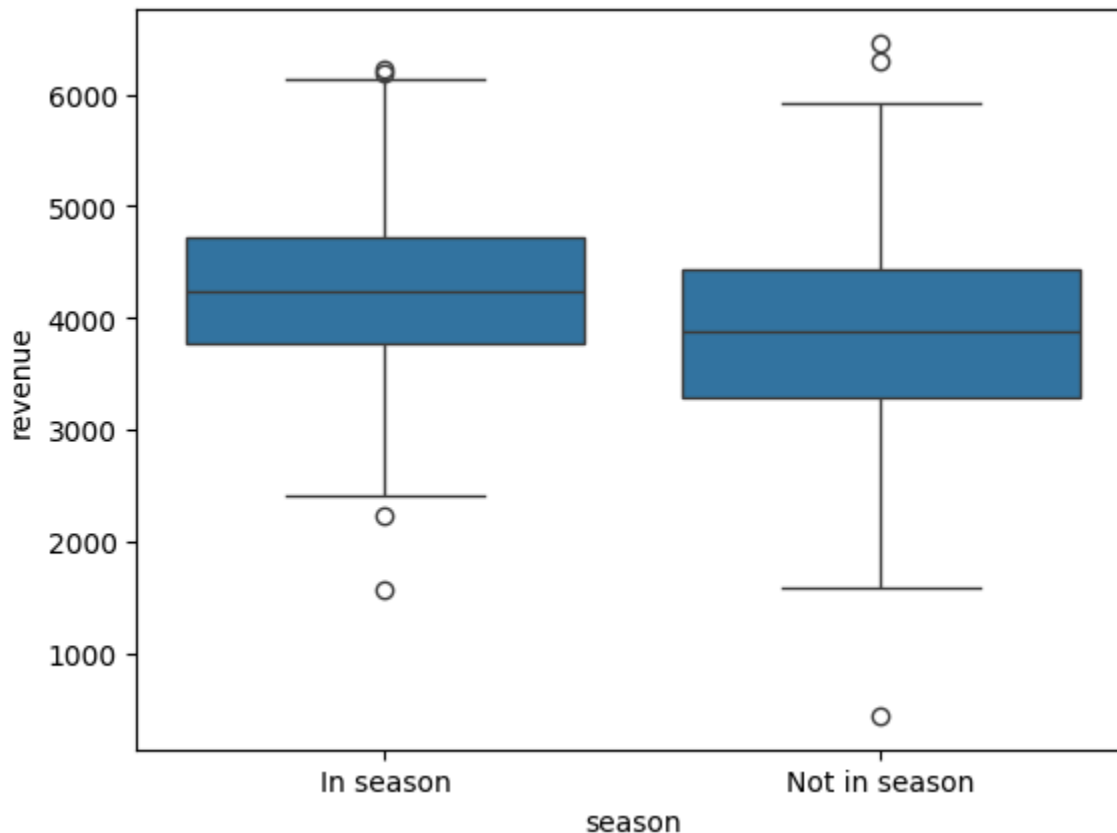
Respiratory visits and revenue(0.89). It proves that respiratory relatedness is the primary driver of total revenue . Any environmental factor that increases these visits will significantly impact the bottom line.

Environmental triggers: pollen_level and respiratory_visits(0.28) there is positive correlation. While 0.28 is “moderate,” it provides the data evidence the client needs. It suggests that as pollen levels rise, respiratory appointments follow.

The “Lag”Effect: pollen_lag1 and pollen_level(0.6) The strong correlation between pollen_level and pollen_lag1 is vital. It suggests that high pollen today predicts high levels(potentially high visits)

Humidity vs Immunity: humid and immunity_visit(0.86) this is a massive standout. High humidity is extremely strongly correlated with immunity - related visits. Marketing for immunity boosts should be aggressively tied to humidity forecasts, not just pollen.

<Axes: xlabel='season', ylabel='revenue'>



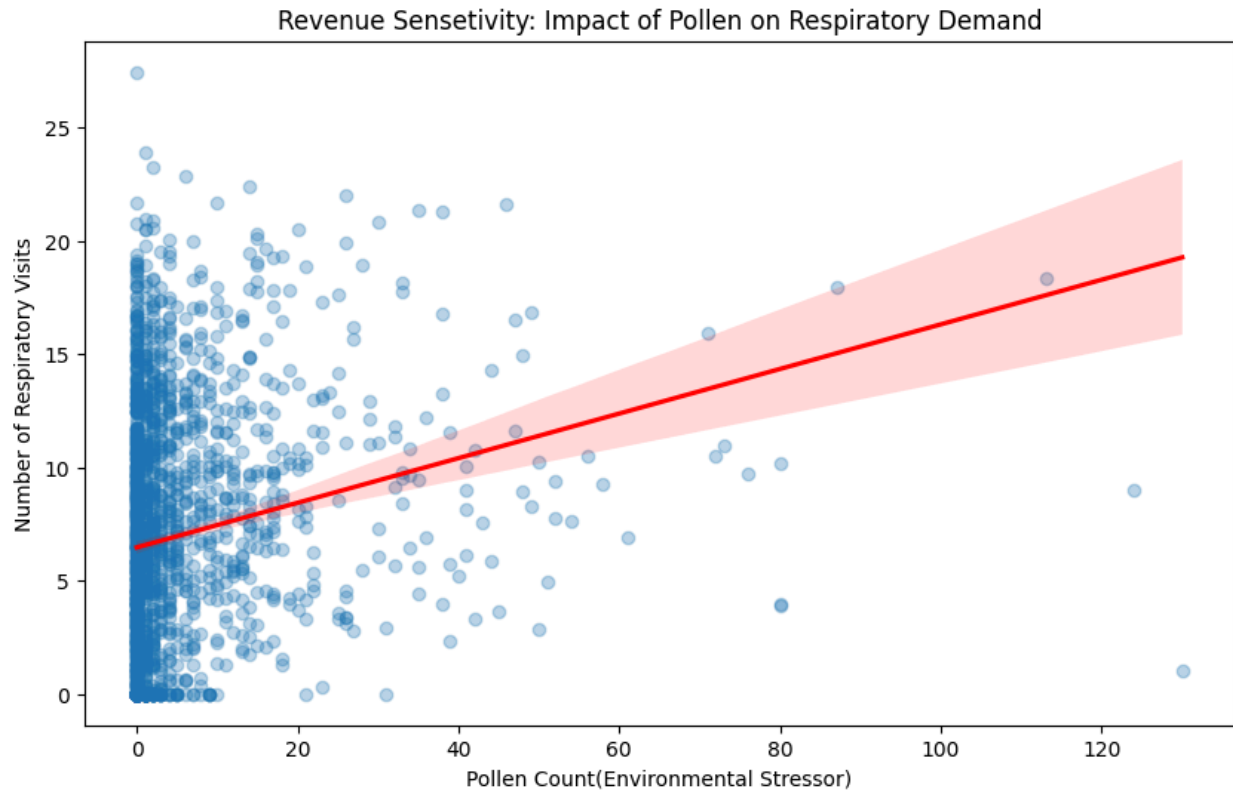
Using the above boxplot we can see how revenue behaves during peak environmental stress periods(In season) versus normal periods(Not in season).

Higher baseline revenue: The median revenue(the horizontal line inside the blue box) for “In season” is visibly higher than “Not In season”. This indicates that, on average, the clinic performs better when environmental triggers(like pollen) are active.

Concentrated consistency: The in season box is slightly more compact and sits higher. This means that during peak environmental periods, revenue is not only higher but also more predictable.

The ceiling effect : The In season period has a higher ceiling , with several days / weeks hitting revenue peaks above 6000.

Revenue Floor: The bottom whisker for “In season” stops significantly higher around 2500 than “Not season” which drops toward 1500. This suggests that environmental factors provide a “safety net” for revenue even on a bad day.



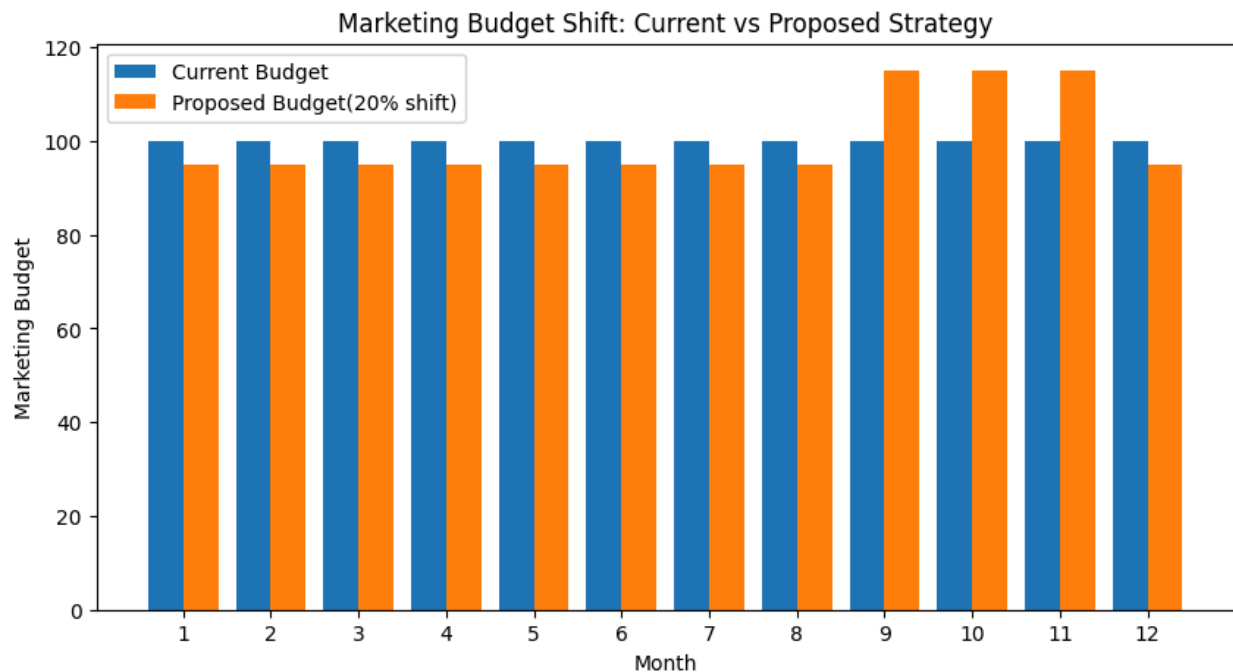
The upward slope of the red line confirms that as the pollen count increases the number of respiratory visits consistently increases as well.

The intercept approx. 6.5 This shows that even when the pollen count is zero, the clinic still expects about 6 to 7 respiratory visits.

The slope of the angle of the red line represents revenue sensitivity. For every unit moved to the right on the x-axis, the line moves up on the y-axis.

Because the entire shaded area is angled upward, the relationship is statistically significant.

Targeted spending: we shouldn't spend marketing dollars equally across the year. The regression shows a clear sensitivity to pollen.



Pollen levels peak in September, October and November makes autumn the most critical period for environmental health risk and clinical demand. Marketing Efforts should therefore be concentrated in these months to maximize patient acquisition and revenue efficiency.

By increasing the budget here we are ensuring Veridian wellness collective is top of mind exactly when local residents are experiencing peak environmental distress.

Budget Neutrality: notice that the orange bars are slightly lower than the blue bars in month 1-8 and 12. This is a shift that we don't ask for more money, but helps to understand how to spend the same money more effectively.

A 20% reallocation of marketing budget toward these peak periods is justified by the observed demand and revenue uplift patterns.

A comparison between current and proposed budget allocation demonstrated improved alignment between marketing spend and environmental demand patterns while maintaining a constant total annual budget.

Conclusion:

Overall, the analysis confirms that the environmental stressors are reliable predictors of healthcare demand, and strategic seasonal marketing and adjustments can improve patient acquisition efficiently and revenue optimization.